

MECH 650 - HW2

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Chapter 2

2.1 Problem 1

We can use Bayes rule to represent the posterior probability as follows:

$$P(z_x = \text{faulty} \mid z_{1:x} < 1) = \frac{P(z_{1:x} < 1 \mid z_x = \text{faulty}) P(z_x = \text{faulty})}{P(z_{1:x} < 1)}$$

By definition, if the sensor is faulty then the measurement will always be less than 1, therefore $P(z_{1:x} < 1 \mid z_x = \text{faulty}) = 1$. However, there is only 1% chance that the sensor is faulty, so $P(z_x = \text{faulty}) = 0.01$.

Now for the denominator term, we can develop using conditional probabilities.

$$\begin{aligned} P(z_{1:x} < 1) &= P(z_{1:x} < 1 \mid z_x = \text{faulty})P(z_x = \text{faulty}) + P(z_{1:x} < 1 \mid z_x = \text{not faulty})P(z_x = \text{not faulty}) \\ &= 1 \cdot 0.01 + \left(\frac{1}{3}\right)^N \cdot 0.99 \end{aligned}$$

So,

$$P(z_x = \text{faulty} \mid z_{1:x} < 1) = \frac{0.01}{0.01 + 0.99 \left(\frac{1}{3}\right)^N}$$

For $N = 10$,

$$P(z_x = \text{faulty} \mid z_{1:10} < 1) = 99.8\%$$

2.2 Problem 3

The solution of this problem relies on the use of Bayes filter to compute the weather distribution recursively.

a) Day 1: We know for a fact that $x_1 = \text{sunny}$

$$bel(x_1) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{matrix} \leftarrow \text{Sunny} \\ \leftarrow \text{Cloudy} \\ \leftarrow \text{Rainy} \end{matrix}$$

Day 2: We can express the state transition matrix as:

$$A = \begin{pmatrix} 0.8 & 0.4 & 0.2 \\ 0.2 & 0.4 & 0.6 \\ 0 & 0.2 & 0.2 \end{pmatrix}$$

Now the predicted state for day 2 can be obtained using:

$$\overline{bel}(x_2) = P(x_2 | x_1) bel(x_1) = A \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.2 \\ 0 \end{pmatrix}$$

Also, the measurement update matrix B_{cloudy} can be written as

$$B_{\text{cloudy}} = \begin{pmatrix} 0.4 & 0 & 0 \\ 0 & 0.7 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Therefore,

$$bel(x_2) = \eta B_{\text{cloudy}} \overline{bel}(x_2) = \eta \begin{pmatrix} 0.32 \\ 0.14 \\ 0 \end{pmatrix}$$

where

$$\eta = \frac{1}{\|(0.32, 0.14, 0)^T\|_1} = 2.17$$

So,

$$bel(x_2) = \begin{pmatrix} 0.7 \\ 0.3 \\ 0 \end{pmatrix}$$

Day 3:

$$\overline{bel}(x_3) = A bel(x_2) = \begin{pmatrix} 0.68 \\ 0.26 \\ 0.06 \end{pmatrix}$$

$$bel(x_3) = \eta B_{\text{cloudy}} \overline{bel}(x_3) = \begin{pmatrix} 0.6 \\ 0.4 \\ 0 \end{pmatrix}$$

Day 4:

$$\overline{bel}(x_4) = A bel(x_3) = \begin{pmatrix} 0.64 \\ 0.28 \\ 0.08 \end{pmatrix}$$

$$bel(x_4) = \eta B_{\text{cloudy}} \overline{bel}(x_4) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Day 5:

$$\overline{bel}(x_5) = A bel(x_4) = \begin{pmatrix} 0.2 \\ 0.6 \\ 0 \end{pmatrix}$$

$$bel(x_5) = \eta B_{\text{cloudy}} \overline{bel}(x_5) = \begin{pmatrix} 0.6 \\ 0.4 \\ 0 \end{pmatrix}$$

b) First, we follow the same method as part (a) to get the posterior beliefs.

$$bel(x_1) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad bel(x_2) = \begin{pmatrix} 0.889 \\ 0.111 \\ 0 \end{pmatrix}, \quad bel(x_3) = \begin{pmatrix} 0.87 \\ 0.13 \\ 0 \end{pmatrix}, \quad bel(x_4) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Then, we apply smoothing. Define the space of possibilities to be $\{S, C, R\}$.

Day 3:

$$P(x_3 = S \mid x_4 = R; z_3 = S) = \frac{P(x_4 = R \mid x_3 = S; z_3 = S)P(x_3 = S \mid z_3 = S)}{P(x_4 = R \mid z_3 = S)} = 0$$

$$P(x_3 = R \mid x_4 = R; z_3 = S) = \frac{P(x_4 = R \mid x_3 = R; z_3 = S)P(x_3 = R \mid z_3 = S)}{P(x_4 = R \mid z_3 = S)} = 0$$

$$P(x_3 = C \mid x_4 = R; z_3 = S) = 1 - 0 - 0 = 1$$

Day 2:

$$P(x_2 = S \mid x_3 = C; z_2 = S) = \frac{P(x_3 = C \mid x_2 = S; z_2 = S)P(x_2 = S \mid z_2 = S)}{P(x_3 = C \mid z_2 = S)}$$

$$= \eta \times 0.2 \times 0.889 = 0.178\eta$$

$$\begin{aligned} P(x_2 = C \mid x_3 = C; z_2 = S) &= \frac{P(x_3 = C \mid x_2 = C; z_2 = S)P(x_2 = C \mid z_2 = S)}{P(x_3 = C \mid z_2 = S)} \\ &= \eta \times 0.4 \times 0.111 = 0.0444\eta \end{aligned}$$

$$P(x_2 = R \mid x_3 = C; z_2 = S) = \frac{P(x_3 = C \mid x_2 = R; z_2 = S)P(x_2 = R \mid z_2 = S)}{P(x_3 = C \mid z_2 = S)} = 0$$

Normalize and compute the probabilities for x_2 :

$$\eta = \frac{1}{0.178 + 0.0444} = 4.496$$

$$P(x_2 = S \mid x_3 = C; z_2 = S) = 0.8$$

$$P(x_2 = R \mid x_3 = C; z_2 = S) = 0$$

$$P(x_2 = C \mid x_3 = C; z_2 = S) = 0.2$$

c) Using the previous part, we can take into account all measurements at every timestep to obtain the sequence Sunny, Sunny, Cloudy, Rainy with

$$p = 1 \times 0.8 \times 1 \times 1 = 0.8$$

Chapter 3

3.1 Problem 1

a) In this problem, the external noise is expressed as a random acceleration input. From newton's second law, we have the following relations tying acceleration to position and velocity:

$$\begin{aligned}\dot{x}_t &= \dot{x}_{t-1} + \ddot{x}_t \Delta t \\ x_t &= x_{t-1} + \dot{x}_{t-1} \Delta t + \ddot{x}_t \frac{\Delta t^2}{2}\end{aligned}$$

Therefore, we need the following state vector $\mathbf{x}_t = \begin{pmatrix} x_t \\ \dot{x}_t \end{pmatrix}$.

b) We can express the above formulas using the state transition matrix A, and the control matrix B:

$$A = \begin{pmatrix} 1 & \Delta t \\ 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} \frac{\Delta t^2}{2} \\ \Delta t \end{pmatrix}$$

Since we know the distribution of the acceleration to be $\mathcal{N}(\mu = 0, \sigma^2 = 1)$, we can directly get R from:

$$R = \text{cov}(B\ddot{x}) = B \text{cov}(\ddot{x}) B^T = \text{Var}(\ddot{x}) B B^T$$

c) From the regular Kalman filter algorithm, the prediction step works as follows:

$$\begin{aligned}\bar{\mu}_t &= A\mu_{t-1} + Bu_t \\ \bar{\Sigma}_t &= A\Sigma_{t-1}A^T + R\end{aligned}$$

Since we know A, B, and R, and $\mu_o = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $u_0 = 0$, we can easily compute the following steps.

t=1:

$$\bar{\mu}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \bar{\Sigma}_1 = \begin{pmatrix} 0.25 & 0.5 \\ 0.5 & 1 \end{pmatrix}$$

t=2:

$$\bar{\mu}_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \bar{\Sigma}_2 = \begin{pmatrix} 2.5 & 2 \\ 2 & 2 \end{pmatrix}$$

t=3:

$$\bar{\mu}_3 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \bar{\Sigma}_3 = \begin{pmatrix} 8.75 & 4.5 \\ 4.5 & 3 \end{pmatrix}$$

t=4:

$$\bar{\mu}_4 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \bar{\Sigma}_4 = \begin{pmatrix} 21 & 8 \\ 8 & 4 \end{pmatrix}$$

t=5:

$$\bar{\mu}_5 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \bar{\Sigma}_5 = \begin{pmatrix} 41.25 & 12.5 \\ 12.5 & 5 \end{pmatrix}$$

d)

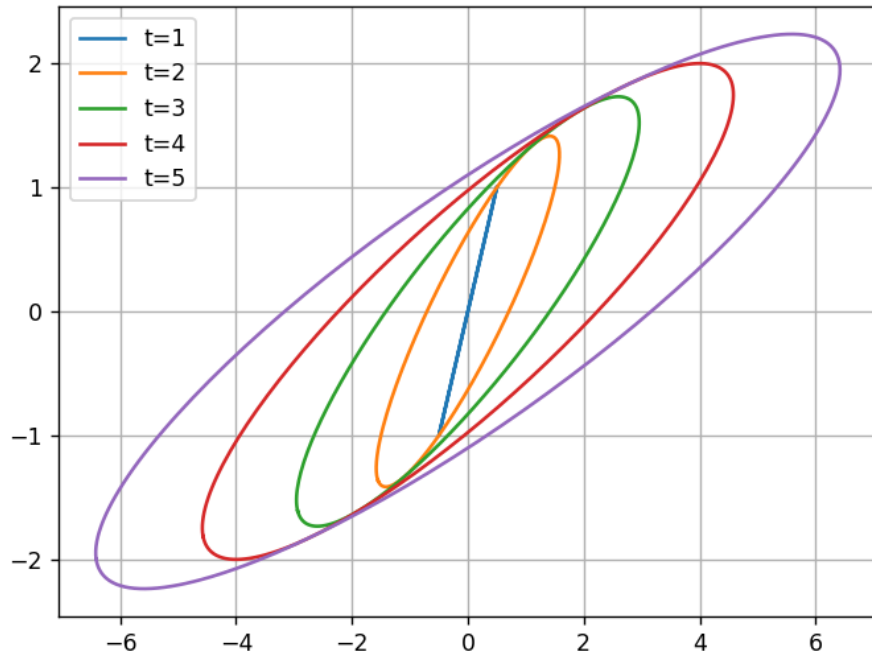


Figure 3.1: Uncertainty ellipses 1 standard deviation away for times 1 through 5.

e) We can notice as $t \rightarrow \infty$, $cov(x, \dot{x}) \rightarrow \approx 0.866$, indicating a strong positive relationship.

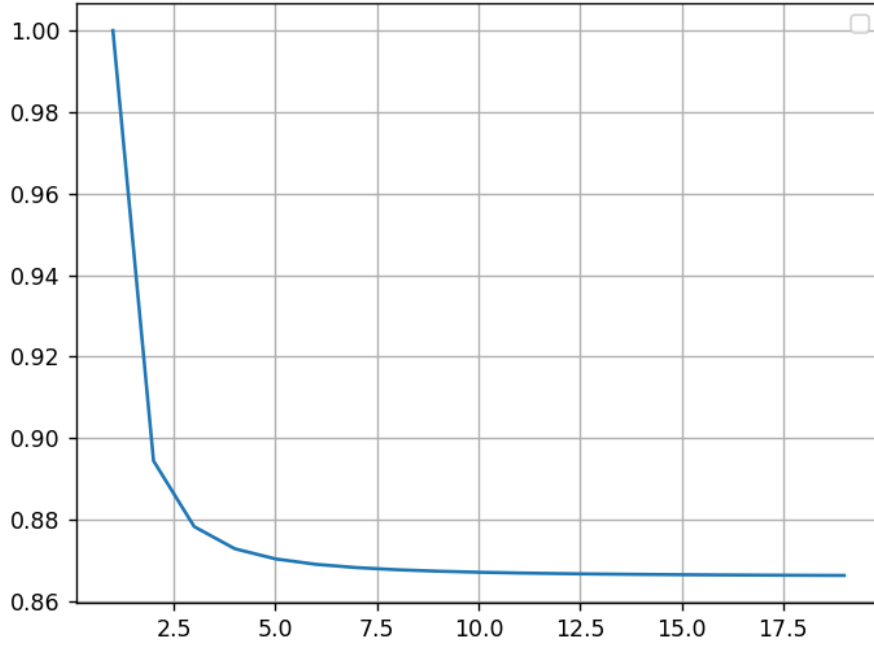


Figure 3.2: Plot of $\text{corr}(x, \dot{x})$ versus t .

3.2 Problem 2

a) I will define the measurement model $z_t = Hx_t$. The covariance matrix Q_t of z_t can be found:

$$Q_t = \text{cov}(Hx_t) = \text{Var}(x_t)HH^T$$

and since the sensor measures only location, then $H = \begin{pmatrix} 1 & 0 \end{pmatrix}$, therefore $Q_t = (\text{Var}(x_t))$.

b) Now, to implement the sensor update, we complete the algorithm of the Kalman filter that we started in Problem 1:

$$K = \bar{\Sigma}H_t^T(H_t\bar{\Sigma}H_t^T + Q_t)^{-1}$$

$$\mu_t = \bar{\mu}_t + K_t(z_t - H_t\bar{\mu}_t)$$

$$\Sigma_t = (I - K_tC_t)\bar{\Sigma}_t$$

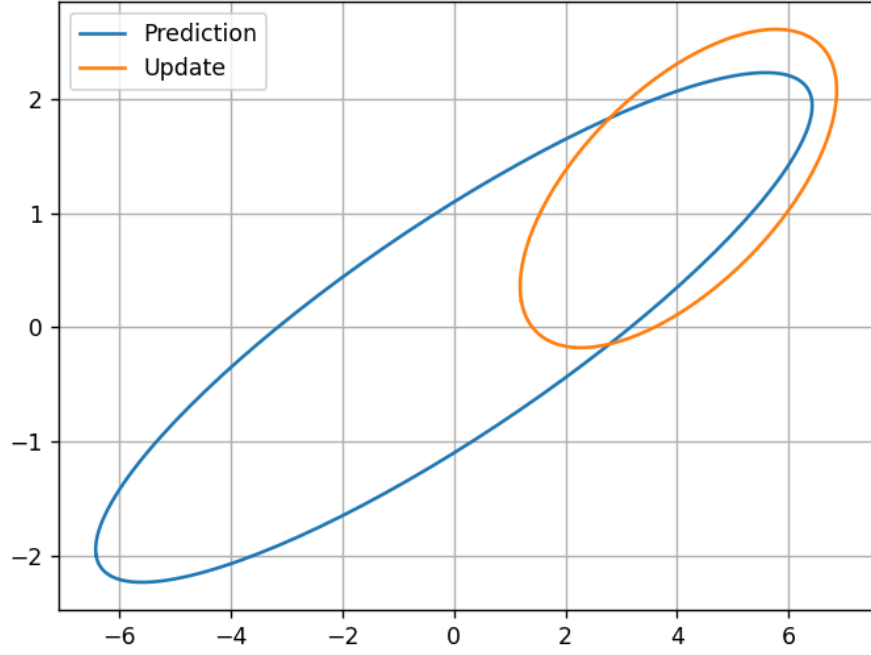


Figure 3.3: Ellipses before and after measurement update.

3.3 Problem 4

a) For this part, I will define the state transition as $A = I_{3 \times 3}$, with state unchanged under 0 control input. As for the control input, the step in the forward direction and its corresponding control matrix are expressed as:

$$u = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad B_t = \begin{pmatrix} \cos(\theta_{t-1}) & 0 & 0 \\ \sin(\theta_{t-1}) & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Now let $x_0 \sim \mathcal{N}(\mu, \Sigma)$, then obtain x_1 through the prediction step $x_1 = Ax_0 + B_1u$. Using a Monte-Carlo simulation, we can observe results clustering around a circle of radius 1 centered at the origin.

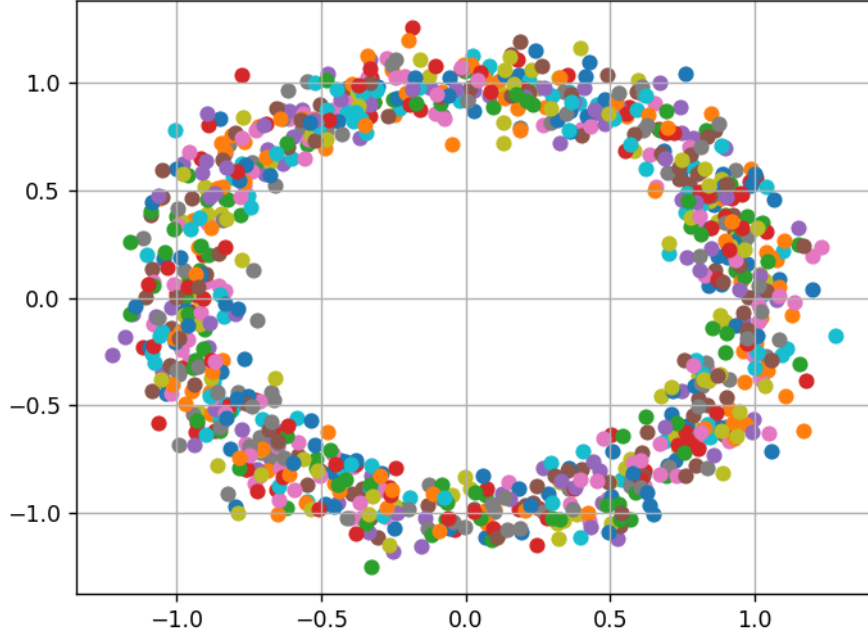


Figure 3.4: Monte-Carlo simulation of the distribution.

b) Let the non-linear function g be such that

$$g(\mathbf{x}_{t-1}) = \begin{pmatrix} x_{t-1} + \cos(\theta_{t-1}) \\ y_{t-1} + \sin(\theta_{t-1}) \\ \theta_{t-1} \end{pmatrix} \quad \text{with } \mathbf{x}_{t-1} = \begin{pmatrix} x_{t-1} \\ y_{t-1} \\ \theta_{t-1} \end{pmatrix}$$

Therefore, we can find the jacobian G of g to be:

$$G_t = \frac{\partial g(\mathbf{x}_{t-1})}{\partial \mathbf{x}_{t-1}} = \begin{pmatrix} 1 & 0 & -\sin(\theta_{t-1}) \\ 0 & 1 & \cos(\theta_{t-1}) \\ 0 & 0 & 1 \end{pmatrix}.$$

The EKF prediction works as follows:

$$\bar{\mu}_t = g(\mu_{t-1}), \quad \bar{\Sigma}_t = G_t \Sigma_{t-1} G_t^T + R_t$$

After replacing with μ_0, Σ_0 , and by taking $R_t = 0$, we obtain:

$$\bar{\mu}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \bar{\Sigma}_1 = \begin{pmatrix} 0.01 & 0 & 0 \\ 0 & 10000.01 & 10000 \\ 0 & 10000 & 10000 \end{pmatrix}$$

c)

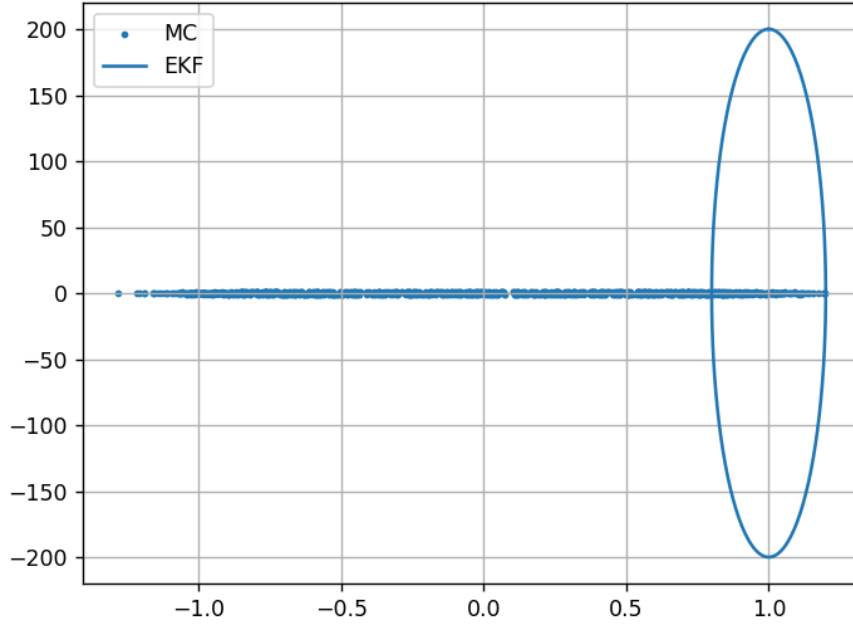


Figure 3.5: Monte-Carlo vs EKF predicted distributions.

Note that the Monte-Carlo simulation gives a circle of radius 1 centered at the origin, while the EKF gives a very tall ellipse centered at $(1, 0)$. This highlights a major failure for the EKF to capture the huge angular uncertainty.

d) Measurement model:

$$z = h(\mathbf{x}) + v, \quad v \sim \mathcal{N}(0, Q)$$

$$h(\mathbf{x}_t) = x_t \quad \Rightarrow \quad H = \frac{\partial h(\mathbf{x}_t)}{\partial \mathbf{x}_t} = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

Given a measurement $z = 0.3$, we can plot the following ring and updated ellipse.

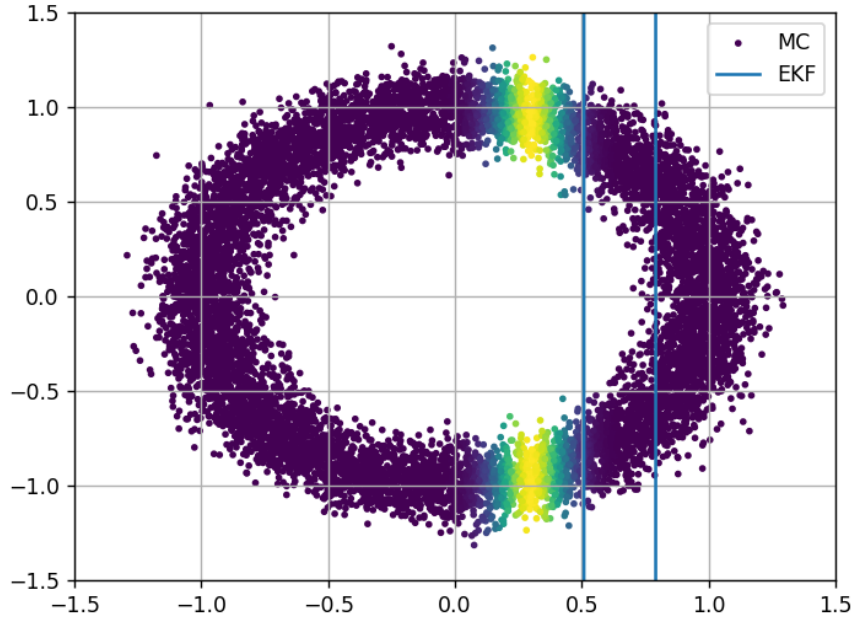


Figure 3.6: Posterior beliefs.

Note that the brightness of the points on the Monte-Carlo distribution indicate probability, with the brightest being the most probable.

Notice in the intuitive posterior, we observe a bimodal distribution with two peaks, something that cannot be captured by a single Gaussian. This is expected, since there are two angles that can amount to $x = 0.3$.

e) My estimate of the posterior makes no assumptions about the data. It is not bounded by linearity. This difference caused the EKF to shift in both the mean and the standard deviations. A considerable upgrade would be to transition to the UKF due to the highly non-linear nature of the problem.

If the uncertainty was in y , the solution of the EKF would have been better considering there is less uncertainty in the non-linearities ($\theta \approx 0$). Therefore $\cos \theta \approx 1$, $\sin \theta \approx 0$. Both methods would predict a vertical blob concentrated around $(1, 0)$.